

Market Microstructure and Price Formation

Mauricio LABADIE
PhD - Quantitative Researcher



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- 1 Efficient Markets, Information Asymmetry and Microstructure
- 2 Price-formation process 1: Kyle
- 3 Price-formation process 2: Glosten-Milgrom
- 4 Conclusions

1 Efficient Markets, Information Asymmetry and Microstructure

2 Price-formation process 1: Kyle

3 Price-formation process 2: Glosten-Milgrom

4 Conclusions

Efficient Market Theory (EMT) I

Postulates of the EMT (1/2)

- **All markets are equal.** They can be considered as a **liquid and frictionless** stock markets, where agents buy and sell assets.
- **All assets are similar and comparable.** They can be fully characterised by the couple risk-return. We can compare via the **Sharpe ratio = return/risk**.
- **There is a unique market price.** All assets have a **fundamental value** p_∞ , which can vary but only exogenously e.g. with news.
- **Information is complete and perfect.** All investors have the same information, e.g. the distribution of the prices and/or returns.

Efficient Market Theory (EMT) II

Postulates of the EMT (2/2)

- **Markets are efficient.** The current price p_t is the best estimate of p_∞ given the current information $\mathcal{F}_t \Rightarrow$ **martingale**.
- **All investors are equal and rational.** They all have the same information and maximise their profits. Only difference is their **utility function**.
- **Rational expectations.** All investors are equally rational and share the same information via p_t and $\mathcal{F}_t \Rightarrow$ **same forecasts**.
- **No endogenous crashes.** Big changes in prices can only be produced externally.

Efficient Market Theory (EMT) III

Critique to the EMT (1/2)

- **All markets are equal.** Markets have transaction costs, regulations and vanishing liquidity. They differ in architecture, geography, creditworthiness ...
- **All assets are similar and comparable.** **Commodities** can be physical or paper. Thousands of liquid **stocks** vs handful of liquid **currencies**.
- **There is a unique market price.** But there are **many prices coexisting**: bid, ask, mid, last trade, average (in time or volume), OHLC (open-high-low-close) ...
- **Information is complete and perfect.** Access to real-time information is expensive and gives a real edge $\Rightarrow$ **Information Asymmetry**.

Efficient Market Theory (EMT) IV

Critique to the EMT (2/2)

- **Markets are efficient.** They can be efficient **in the long run but not instantaneously**: they need **time to digest** new information.
- **All investors are equal and rational.** Investors have different investment styles or biases, and sometimes they have herd behaviour.
- **Rational expectations.** An investor can choose with the heart, not with the head: iPhone or Android? BMW or Mercedes Benz? Red or blue shirt?
- **No endogenous crashes.** Black Monday and portfolio insurance (Oct 19 1987)? Flash Crash and HFT (May 6 2010)? Subprimes (2007)?

Information Asymmetry I

Description

Joseph Stiglitz won the Nobel Price in Economics in 2001 for his work on Information Asymmetry:

- According to the EMT, information is complete and perfect.
- But in reality, **information is not homogeneously distributed**:
 - ★ **Yahoo! Finance**: free but 15-minute delay.
 - ★ **Bloomberg**: 25k USD/year but real-time quotes and news.
- Even if information were public and free, there are **asymmetries everywhere**:
 - ★ **Sources**: Reuters, Financial Times, evening news, etc.
 - ★ **Information processing**: C++, Java, R, Matlab, Excel, etc.
 - ★ **Analysis**: macroeconomics, accounting, statistics, time series, etc.

Information Asymmetry II

Principal hypothesis

There are **two types of traders or investors**: informed and non-informed traders.

- **Informed traders:**

- ★ They have an informational advantage and use it for profit.

⇒ **Directional traders.**

- ★ They know the fundamental price of an asset e.g. **Kyle model**.

- ★ They know if prices will go up or down based on some private news e.g. **Glosten-Milgrom model**.

- **Non-informed traders:**

- ★ They have no information on fundamental prices.

- ★ They trade in any direction, based on **non-fundamentals** e.g. hedging, brokerage, etc.

- ★ Their actions hide the trading of informed traders.

⇒ **Noise traders.**

Information Asymmetry III

Information gives an edge

- Access to data:
 - ★ More information $\Rightarrow$ **decision-making process more robust.**
 - ★ High-frequency traders pay to have **faster, cleaner data** at the millisecond scale.
 - Knowledge and inference:
 - ★ **More reliable inferences** give an advantage: Kyle and Glosten-Milgrom.
 - ★ **Brain power:** Access to information could be free, but mining and processing it faster gives an advantage.
- $\Rightarrow$ **Brain drain**, highly-skilled individuals (PhDs, MBAs, Masters) in Finance.

Information Asymmetry IV

Insider trading

- It is trading a **public company's** stock with **private information**:
 - ★ You are aware of a crucial event before it goes public: new product release, CEO resign, merger, bankruptcy, etc.
 - ★ You know the quarterly results before the conference call with investors.
- It is **illegal** because it is considered as **unfair**:
 - ★ It hurts investors who ignore the private information.
 - ★ It can hurt the reputation of a firm.
 - ★ At least 93 countries have laws against insider trading.
- But there is a case **supporting insider trading**:
 - ★ Compromise between preserving incentives for innovation and maintaining accurate securities pricing.
 - ★ It is an efficient compensation scheme: corporate entrepreneurs and performance bonuses.
 - ★ It is hard to detect and punish, so why waste money on it?

Market Microstructure I

Definition

- Harris 2002: “It is the branch of financial economics that investigates **trading and the organisation of markets**”.
 - Each market, asset and investor is considered unique.
 - Different, almost opposite approach of EMT:
 - ★ **EMT** is axiomatic, deductive, **top-down**.
 - ★ **Microstructure** is empirical, inductive, **bottom-up**.
 - Analogy with **Newton**’s Natural Philosophy and Sciences:
 - ★ Empirical, intuitive approach, avoiding a priori assumptions.
 - ★ Mechanisms and analogies rather than axioms and theorems.
 - ★ General principles emerge from particular cases sharing same behaviour.
- ⇒ **Out of mainstream Economics.**
- ⇒ **New approaches:** Statistics, Econophysics, Computer Science, Evolution, etc.

Market Microstructure II

Microstructure in detail

Heterogeneous factors have to be taken into account, case by case:

- **Market design**: floor vs electronic markets, priority/matching rules, trading times, lunch breaks, etc.
- **Agents**: brokers, dealers, market-makers, informed traders, noise traders, arbitrageurs, etc.
- **Transaction costs**: commissions, fees, spreads, market impact, market risk, etc.
- **Liquidity**: fluctuations in volume, fragmented markets, dark pools, etc.
- **Benchmarks**: open, close, OHLC (open-high-low-close), VWAP (volume-weighted average price), etc.
- **Prices**: bid, ask, mid, last traded, average (in time or in volume), etc.

Market Microstructure III

Crucial concepts in Microstructure

- **Feedback:**

- ★ The structure and dynamics of markets influence the behaviour of agents.
- ★ Every single action of every single agent influences the market.
- ⇒ Feedback loop **market** ↔ **agent**.
- ★ In EMT, agents are infinitesimal ⇒ no feedback.

- **Emergence:**

- ★ From all interactions **market** ↔ **agent** and **agent** ↔ **agent**, new properties appear.
- ★ Add or remove an agent or a market feature ⇒ old properties can disappear and new ones appear.
- ⇒ **Systems theory**: the whole is always bigger than the sum of its parts.

- **Meta-stability:**

- ★ Two opposite forces keep things in equilibrium: **Butterfly effect vs market resilience**.
- ★ Small variations ⇒ the system's equilibrium does not change.
- ★ But with big changes old equilibria can disappear and new equilibria arise.

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Kyle Model (1985). Description I

Hypotheses

- The model proposes an **iterative price-formation process**.
- There are two types of traders:
 - ★ **One informed trader**, who knows the fundamental price p_0 of the asset.
 - ★ **Several non-informed traders** or noise traders, who either ignore p_0 or trade regardless of price (e.g. hedgers).
- There is **one dealer**, who knows the law $\tilde{v} \sim \mathcal{N}(p_0, \Sigma_0)$ but ignores the fundamental value p_0 .
- The non-informed traders, who completely ignore $\tilde{v}$, buy a quantity $\tilde{u} \sim \mathcal{N}(0, \sigma_u^2)$ of shares.
- The informed trader knows p_0 and buys a quantity $\tilde{x}$ of shares.
- The dealer observes the net aggregated volume $x + u$ but does not know the values x and u separately.

Kyle Model (1985). Description II

Dealer's price

It is a linear function of the aggregated volume she observes, i.e.

$$P(x + u) = \mu + \lambda(x + u), \quad (1)$$

where

- λ is the market impact, i.e. the amount in EUR the dealer changes her price μ per traded share.
- $1/\lambda$ represents the depth of the market, i.e. the number of traded shares needed to push the price by 1 EUR.
- The market impact is assumed linear.

Informed trader's profit

$$\mathbb{E}[(\tilde{v} - P(x + \tilde{u}))x \mid \tilde{v} = v] = (v - \mu)x - \lambda x^2. \quad (2)$$

Kyle Model (1985). Equilibrium I

Optimal shares for the informed trader

For a unique maximum of (2) we assume $\lambda > 0$. The value of x that maximises the informed trader's profit is thus

$$X(v) = \beta(v - \mu), \quad \beta := \frac{1}{2\lambda}.$$

Projection Theorem

If $Y = (Y_1, Y_2)'$ is a bivariate Normal random variable of mean and covariance matrix

$$m_Y = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$$

then

$$\mathbb{E}[Y_1 | Y_2] = m_1 + \frac{\sigma_{12}}{\sigma_2^2}(Y_2 - m_2). \quad (3)$$

Kyle Model (1985). Equilibrium II

Application of the Projection Theorem

For the vector $Y = (\tilde{v}, \tilde{x} + \tilde{u})'$, one can show that

$$m_Y = \begin{pmatrix} p_0 \\ \beta(p_0 - \mu) \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_0 & \beta\Sigma_0 \\ \beta\Sigma_0 & \sigma_u^2 + \beta^2\Sigma_0 \end{pmatrix}.$$

Therefore, the Projection Theorem implies that

$$\mathbb{E}[\tilde{v}|\tilde{x} + \tilde{u}] = p_0 + \frac{\beta\Sigma_0}{\sigma_u^2 + \beta^2\Sigma_0}(\tilde{x} + \tilde{u} - \beta(p_0 - \mu)). \quad (4)$$

Kyle Model (1985). Equilibrium III

Optimal values of λ , β and μ

In an efficient market we have

$$P(\tilde{x} + \tilde{u}) = \mathbb{E}[\tilde{v} | \tilde{x} + \tilde{u}]$$

for every μ , $\tilde{x}$ and $\tilde{u}$.

- If we choose $\mu = p_0$ and $\tilde{x} + \tilde{u} = 1$ then

$$\lambda = \frac{\beta \Sigma_0}{\sigma_u^2 + \beta^2 \Sigma_0} = \frac{1}{2} \left(\frac{\Sigma_0}{\sigma_u^2} \right)^{1/2}, \quad \beta = \frac{1}{2\lambda} = \left(\frac{\sigma_u^2}{\Sigma_0} \right)^{1/2}.$$

- From above we have

$$\mu + \lambda(\tilde{x} + \tilde{u}) = p_0 + \lambda(\tilde{x} + \tilde{u} - \beta(p_0 - \mu)).$$

Therefore, if we choose $\tilde{x} = \tilde{u} = 0$ then

$$\mu - p_0 = \lambda\beta(\mu - p_0).$$

But since $\lambda\beta = 1/2$ we necessarily have that $\mu = p_0$.

Kyle Model (1985). Equilibrium IV

Kyle's equilibrium

$$\begin{aligned}P(\tilde{x} + \tilde{u}) &= p_0 + \lambda(\tilde{x} + \tilde{u}) \\X(\tilde{v}) &= \beta(\tilde{v} - p_0), \\ \beta &:= (\sigma_u^2 / \Sigma_0)^{1/2}, \\ \lambda &:= (1/2) (\Sigma_0 / \sigma_u^2)^{1/2}.\end{aligned}\tag{5}$$

Remarks I

Optimal shares and uncertainty

The optimal number of shares to buy for the informed trader is

$$x = \beta(v - p_0) = \left(\frac{\sigma_u^2}{\Sigma_0} \right)^{1/2} (v - p_0)$$

and her profit $\tilde{\pi}$ is

$$\tilde{\pi} = (v - p_0)x - \lambda x^2 = \frac{\beta}{2} (v - p_0)^2.$$

Therefore,

$$\mathbb{E}[\tilde{\pi}] = \frac{1}{2} (\sigma_u^2 \Sigma_0)^{1/2}.$$

- If σ_u^2 increases then x and $\tilde{\pi}$ increase: more profit and hiding in the crowd.
- If Σ_0 decreases then x increases but $\tilde{\pi}$ decreases: losing stealth, acting faster.

Remarks II

At each iteration, the price variance gets reduced

$$\begin{aligned}\text{Var}(\tilde{p}) &= \lambda^2 \text{Var}(\tilde{x} + \tilde{u}) \\ &= \frac{1}{4} \frac{\Sigma_0}{\sigma_u^2} (\beta^2 \Sigma_0 + \sigma_u^2) \\ &= \frac{1}{4} \frac{\Sigma_0}{\sigma_u^2} (2\sigma_u^2) \\ &= \frac{1}{2} \Sigma_0 .\end{aligned}$$

⇒ With her trade, the informed trader reveals half of her information to the dealer.

Remarks III

Asymptotics of the dealer's variance

Recursively, one can show that

$$\text{Var}(\tilde{p}_n) = \left(\frac{1}{2}\right)^n \Sigma_0 .$$

Therefore,

$$\lim_{n \rightarrow \infty} \text{Var}(\tilde{p}_n) = 0 .$$

Remarks IV

Asymptotics of the dealer's price

Let p_∞ be the fundamental value, known by the informed trader. After n iterations the dealer's price is

$$\begin{aligned} p_n &= p_{n-1} + \lambda(\tilde{x} + \tilde{u}) \\ &= p_{n-1} + \frac{1}{2}(p_\infty - p_{n-1}) + \lambda\tilde{u}. \end{aligned}$$

Rearranging we obtain

$$p_n - p_{n-1} = -\frac{1}{2}(p_{n-1} - p_\infty) + \lambda\tilde{u}.$$

Solving the associated ODE

$$dp = -\frac{1}{2}(p - p_\infty) + \lambda\tilde{u}$$

yields

$$p(t) = p_\infty + (p(0) - p_\infty - 2\lambda\tilde{u})e^{-t/2} + 2\lambda\tilde{u}. \quad (6)$$

Remarks V

The dealer's price converges to the fundamental price

- Taking expectations on (6) and using $\mathbb{E}[\tilde{u}] = 0$ we obtain

$$\mathbb{E}[p(t)] = p_{\infty} + (p(0) - p_{\infty})e^{-t/2}.$$

In consequence $\mathbb{E}[p(t)] \rightarrow p_{\infty}$.

- Recall that $\lambda = (1/2)(\Sigma_0/\sigma_u^2)^{1/2}$. After one step we have $\Sigma_1 = \Sigma_0/2$, which implies that

$$\lambda_1 = \left(\frac{1}{2}\right)^{1/2} \lambda_0, \quad \lambda_0 := (1/2)(\Sigma_0/\sigma_u^2)^{1/2}.$$

Recursively we obtain that

$$\lambda_n = \left(\frac{1}{2}\right)^{n/2} \lambda_0.$$

In consequence $\lambda_n \rightarrow 0$, which implies that $p(t) \rightarrow p_{\infty}$.

Remarks VI

Market efficiency

- The market is **asymptotically efficient**: the dealer's price p_t converges to the fundamental price p_∞ .
 - But it is **not instantaneously efficient**: the dealer needs several trading rounds to hit the fundamental price.
- ⇒ Market needs time to digest new information.
- Thanks to informed traders, the market is efficient in the long run.
 - But in the short run, informed traders can make substantial profits.
- ⇒ $\alpha \neq 0$ is not in contradiction with market efficiency.

Remarks VII

The model does not require a fundamental price

- The dealer knows that the informed trader knows the fundamental price p_∞ .
 - Every iteration, the dealer extracts information from the informed trader's flow and her price gets closer to p_∞ .
 - This holds for any price the informed trader chooses: **the dealer's price converges to whatever price $p^\#$ the informed trader thinks is fair.**
 - It does not matter if $p^\# \neq p_\infty$ or even if p_∞ does not exist at all.
- ⇒ **The informed trader drives the price-formation process.**
- ⇒ Financial bubbles and crashes are not as exogenous as the EMT claims.

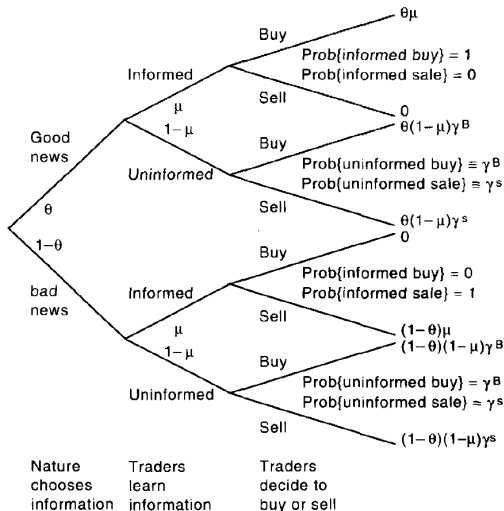
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Glosten-Milgrom Model (1985). Description I

Hypotheses

- The model tries to explain how the dealer updates her price given that the last order she received was a sell or a buy.
- As in the Kyle model, we have several informed traders, several non-informed traders and one dealer.
- At time $t = 0$ the asset V has a price V_0 .
- At time $t = 0$ there are news: good news with probability θ and bad news with probability $1 - \theta$.
- Good news $\Rightarrow$ new price is $V^+ > V_0$; bad news $\Rightarrow$ new price is $V^- < V_0$.
- Everybody knows the news' effect, but only informed traders know if the news are good or bad.
- At time $t = 0$ the dealer receives an order, either a buy $Q = B$ or a sell $Q = S$.
- She does not know if the order comes from an informed or a non-informed trader, but she knows the probabilities: μ for informed, $1 - \mu$ for non-informed.
- If good (resp. bad) news the informed traders buy (resp. sell) with probability 1.
- Non-informed traders always buy with probability γ^B and sell with probability γ^S .

Glosten-Milgrom Model (1985). Description II



Glosten-Milgrom Model (1985). Solution I

Dealer's price updates

At time $t = 1$ the dealer updates her selling (ask) price a_1 and her buying (bid) price b_1 as a function of the last order Q she received:

$$a_1 = \mathbb{E}[V \mid Q = B] = V^+ \mathbb{P}[V = V^+ \mid Q = B] + V^- \mathbb{P}[V = V^- \mid Q = B], \quad (7)$$

$$b_1 = \mathbb{E}[V \mid Q = S] = V^+ \mathbb{P}[V = V^+ \mid Q = S] + V^- \mathbb{P}[V = V^- \mid Q = S].$$

To obtain a_1 and b_1 we **only** need to compute the probabilities

$$\mathbb{P}[V = V^\pm \mid Q = B, S].$$

Bayes' Theorem

For any two events X and Y we have

$$\mathbb{P}[X \mid Y] = \frac{\mathbb{P}[Y \mid X] \mathbb{P}[X]}{\mathbb{P}[Y]}.$$

Glosten-Milgrom Model (1985). Solution II

Application of Bayes' Theorem

We first compute $\mathbb{P}[V = V^+ | Q = B]$:

$$\begin{aligned}\mathbb{P}[V^+ | B] &= \frac{\mathbb{P}[B | V^+] \mathbb{P}[V^+]}{\mathbb{P}[B]} \\ &= \frac{\mathbb{P}[B | V^+] \mathbb{P}[V^+]}{\mathbb{P}[B | V^+] \mathbb{P}[V^+] + \mathbb{P}[B | V^-] \mathbb{P}[V^-]}.\end{aligned}\tag{8}$$

Each element of (8) is computed using the decision tree:

- $\mathbb{P}[V^+] = \theta$.
- $\mathbb{P}[V^-] = 1 - \theta$.
- $\mathbb{P}[B | V^+] = \mu \times 1 + (1 - \mu) \times \gamma^B$.
- $\mathbb{P}[B | V^-] = \mu \times 0 + (1 - \mu) \times \gamma^B$.

Similarly, we compute

$$\mathbb{P}[V^+ | S], \quad \mathbb{P}[V^- | B] \quad \text{and} \quad \mathbb{P}[V^- | S].$$

With these four conditional probabilities, the dealer can update her quotes using (7).

Numerical solution for $t = 1$ I

Preparing Bayes' bazooka

Suppose $V^+ = 100$, $V^- = 20$ and $\theta = \mu = \gamma^B = \gamma^S = 1/2$.

- At $t = 0$ the dealer's price is

$$\mathbb{E}[V] = (100)(1/2) + (20)(1/2) = 60.$$

- At $t = 1$ we compute the direct conditional probabilities:

- ★ $\mathbb{P}[V^+] = \mathbb{P}[V^-] = 1/2$.
- ★ $\mathbb{P}[B|V^+] = \mathbb{P}[S|V^-] = (1/2)(1) + (1/2)(1/2) = 3/4$.
- ★ $\mathbb{P}[S|V^+] = \mathbb{P}[B|V^-] = (1/2)(0) + (1/2)(1/2) = 1/4$.

Numerical solution for $t = 1$ II

Shooting Bayes' bazooka

At $t = 1$ we infer via Bayes' Theorem the reverse conditional probabilities:

$$\begin{aligned}\mathbb{P}[V^+ | B] &= \frac{\mathbb{P}[B|V^+] \mathbb{P}[V^+]}{\mathbb{P}[B|V^+] \mathbb{P}[V^+] + \mathbb{P}[B|V^-] \mathbb{P}[V^-]} \\ &= \frac{(3/4)(1/2)}{(3/4)(1/2) + (1/4)(1/2)} = 3/4,\end{aligned}$$

$$\begin{aligned}\mathbb{P}[V^- | B] &= \frac{\mathbb{P}[B|V^-] \mathbb{P}[V^-]}{\mathbb{P}[B|V^+] \mathbb{P}[V^+] + \mathbb{P}[B|V^-] \mathbb{P}[V^-]} \\ &= \frac{(1/4)(1/2)}{(3/4)(1/2) + (1/4)(1/2)} = 1/4.\end{aligned}$$

$$\mathbb{P}[V^+ | S] = 1/4, \quad \mathbb{P}[V^- | S] = 3/4.$$

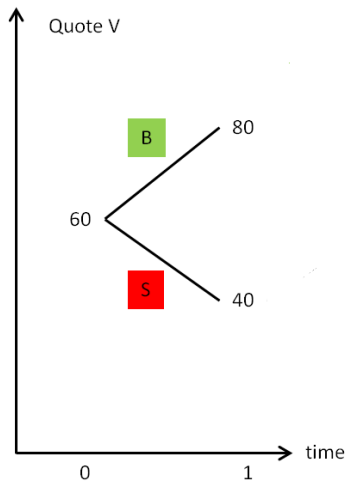
Numerical solution for $t = 1$ III

The dealer updates her quotes

$$\begin{aligned}a_1 &= \mathbb{E}[V \mid B] \\&= V^+ \mathbb{P}[V^+ \mid B] + V^- \mathbb{P}[V^- \mid B] \\&= (100)(3/4) + (20)(1/4) = 80.\end{aligned}$$

$$\begin{aligned}b_1 &= \mathbb{E}[V \mid S] \\&= V^+ \mathbb{P}[V^+ \mid S] + V^- \mathbb{P}[V^- \mid S] \\&= (100)(1/4) + (20)(3/4) = 40.\end{aligned}$$

Numerical solution for $t = 1$ IV



Numerical solution for $t = 2$ I

Preparing Bayes' bazooka

Suppose that at time $t = 1$ the dealer received a buy, i.e. $Q_1 = B$. From the previous step we computed

$$\mathbb{P}[V^+|B_1] = 3/4, \quad \mathbb{P}[V^-|B_1] = 1/4,$$

Since the decision tree is valid for all times, we find that Q_1 and Q_2 are independent. Therefore,

$$\mathbb{P}[B_2|V^+] = \mathbb{P}[S_2|V^-] = 3/4,$$

$$\mathbb{P}[S_2|V^+] = \mathbb{P}[B_2|V^-] = 1/4.$$

Numerical solution for $t = 2$ II

Shooting Bayes' bazooka given $Q_1 = B$

$$\begin{aligned}\mathbb{P}[V^+ | B_2, B_1] &= \frac{\mathbb{P}[B_2 | V^+] \mathbb{P}[V^+ | B_1]}{\mathbb{P}[B_2 | V^+] \mathbb{P}[V^+ | B_1] + \mathbb{P}[B_2 | V^-] \mathbb{P}[V^- | B_1]} \\ &= \frac{(3/4)(3/4)}{(3/4)(3/4) + (1/4)(1/4)} = 9/10.\end{aligned}$$

$$\begin{aligned}\mathbb{P}[V^- | B_2, B_1] &= \frac{\mathbb{P}[B_2 | V^-] \mathbb{P}[V^- | B_1]}{\mathbb{P}[B_2 | V^+] \mathbb{P}[V^+ | B_1] + \mathbb{P}[B_2 | V^-] \mathbb{P}[V^- | B_1]} \\ &= \frac{(1/4)(1/4)}{(3/4)(3/4) + (1/4)(1/4)} = 1/10.\end{aligned}$$

$$\mathbb{P}[V^+ | S_2, B_1] = \mathbb{P}[V^+ | B_2, S_1] = 1/2.$$

Numerical solution for $t = 2$ III

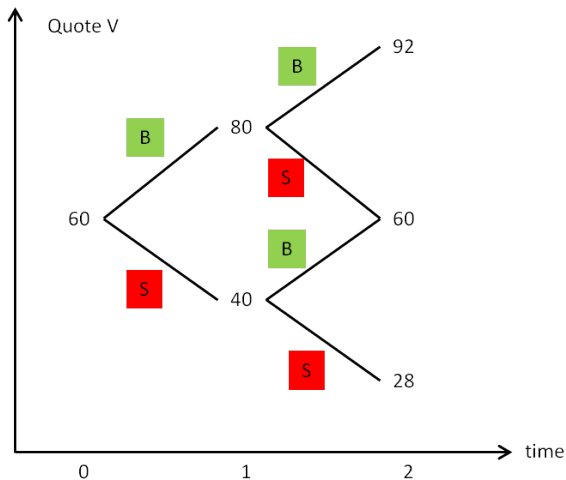
The dealer updates her quotes given $Q_1 = B$

$$\begin{aligned}a_2(Q_1 = B) &= \mathbb{E}[V \mid B_2, B_1] \\&= (100)(9/10) + (20)(1/10) = 92. \\b_2(Q_1 = B) &= \mathbb{E}[V \mid S_2, B_1] \\&= (100)(1/2) + (20)(1/2) = 60.\end{aligned}$$

The dealer updates her quotes given $Q_1 = S$

$$\begin{aligned}a_2(Q_1 = S) &= \mathbb{E}[V \mid B_2, S_1] \\&= (100)(1/2) + (20)(1/2) = 60. \\b_2(Q_1 = S) &= \mathbb{E}[V \mid S_2, S_1] \\&= (100)(1/10) + (20)(9/10) = 28.\end{aligned}$$

Numerical solution for $t = 2$ IV



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Final comments

Summary of this presentation

- We described and challenged the Efficient Market Theory (EMT).
- We saw what is Information Asymmetry and how it tackles some of the “defaults” of the EMT.
- We compared the approach of Microstructure with that of the EMT.
- We understood the price-formation process (Kyle and Glosten-Milgrom):
 - ★ Dealers trying to infer price information from trading flow.
 - ★ Informed traders trying to hide their intentions behind noise traders.
 - ★ Every trading round leaks information from informed traders to dealers.
 - ★ Markets are efficient: asymptotically, not instantaneously.
 - ⇒ Hide-and-seek game between dealers and informed traders.
- We had a first glimpse of market impact:
 - ★ Buying (selling) pushes prices up (down).
 - ★ The more you trade, the more you push market prices against you.
 - ★ Linear, although in reality it is nonlinear: concave.

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